ZentroMart AI Voice Support Agent

Evidence-grounded support | multi-turn workflows | Gnani.ai speech

REFERENCE DATE | 20 OCT 2026

OFFLINE TESTED

LIVE API TESTS PENDING

01 Problem and solution

ZentroMart support handles repeated questions about delivery, cancellations, returns, refunds, warranty and damaged goods. This capstone implements an AI support agent that retrieves company evidence, keeps conversation context, gates protected order lookups behind caller verification, and uses the same agent for text and voice.

02 Core capabilities

Evidence-grounded answers

Section-aware Markdown ingestion; source/effective-date metadata; policy precedence; abstention when evidence is absent.

Verified order support

Uses supplied lookup tool; asks for phone last four or delivery pincode; no order facts before tool verification.

Human handover

Uses the provided ticket tool for configured escalation rules; confirms a ticket only after `created=True`.

Voice interface

Browser microphone Record / Stop / preview / Send Audio, plus upload; actual audio is converted to WAV and sent through Gnani STT -> same agent -> Gnani TTS. Real-mic regression still pending.

03 Technology stack

Python | FastAPI | TF-IDF + cosine | OpenAI-compatible LLM | Gnani.ai STT / TTS

IMPLEMENTATION STATUS

45 automated tests pass, dataset validation passes, and retrieval finds 81/90 gold evidence IDs in the top five. Live baseline/RAG answer scoring and all 96 conversation turns remain pending without provider keys. A prior uploaded-audio STT/TTS pass was reported; browser-microphone end-to-end testing remains pending on Windows.

Architecture and implementation

One shared agent for text and voice

ONE TURN / END-TO-END PIPELINE

DATA PLANE 21 KB Markdown documents + 70 product rows | 7 database CSVs | holidays.csv | supplied `lookup_order` and `create_ticket` tools | practice questions and conversations

04 Implementation details

Ingestion and chunking

Reads front matter, splits on `[SECTION-ID]` headings and paragraph boundaries, and preserves source, status, effective date, version and chunk index.

Retrieval and authority

TF-IDF n-gram vectors + cosine similarity; current policies receive priority and a per-source diversity cap. Superseded return text is conditional-only for the legacy case preserved by current RET-3.

Session and protected data

Session state retains recent turns and order context. The supplied tool enforces proof matching; contact fields and pincode proof are filtered from model context.

Escalation and tool truth

Order lookup, business-day checks and ticket actions are deterministic code. Ticket ID/handover wording depends on the actual tool result; failed creation is reported as unconfirmed.

SPEECH INTEGRATION BOUNDARY

Official `gnani-vachana` adapters are configured for STT `en-IN` and TTS `timbre-v2.5` / `Kaveri`. Browser `MediaRecorder` output is decoded and encoded as true WAV before upload. No alternate speech provider is used. The package's changed microphone workflow still requires a real Windows-device + live-Gnani check; no credentials are available in this build workspace.

Evaluation and verified results

Offline metrics only where measured

45 / 45
pytest tests passed

90.0%
gold evidence Recall@5

47 / 47
questions hit at least one gold item

0
live API calls claimed

RESULTS / STATUS

Metric	Measured result	Interpretation
Dataset validator	PASS	9 CSVs, 21 KB docs, keys, order math, practice evidence and verification gates checked.
Automated test suite	45 passed / 0 failed	pytest run with offline mocks and supplied-data integration tests.
Gold evidence Recall@5	81/90 = 90.0%	Exact gold section/source IDs found in top-five retrieved chunks.
Question hit rate@5	100% (47/47)	At least one gold evidence item retrieved for each question with a gold label.
Step 1 baseline accuracy	PENDING	Requires configured live LLM; no answer accuracy inferred from retrieval alone.
Step 2 RAG answer accuracy	PENDING	Same 52 questions and rubric await the live LLM evaluation.
Conversation evaluation	0 / 96 turns executed	Live evaluation status; answer and context checks require an LLM-configured run.
Gnani STT / TTS	UPLOAD TURN REPORTED PASS	A prior local uploaded-audio transcript and audible reply were reported; updated microphone flow still needs a real Windows/Gnani run.
Application / batch smoke	PASS (offline)	`/` and `/health`: HTTP 200; no-key chat: 503 as expected. No live provider was called in this packaging workspace.

HOW TO READ THE RETRIEVAL SCORE

Recall@5 = the fraction of supplied gold evidence IDs matched against the top-five retrieved source/section metadata. The 100% question hit rate means at least one gold item was found for each of 47 labelled questions. Neither metric proves the generated answer is correct. Product queries with unspecified variants can remain ambiguous; baseline/RAG correctness, unsupported-answer rate and conversation pass rate remain unmeasured.

Limitations and reproduction

Remaining steps before final submission

05 Limitations and next work

Live scores and new microphone path pending.

No provider keys were available in this packaging workspace. Run the full baseline/RAG and 96-turn evaluations, then test a real browser microphone recording through live Gnani STT/TTS on Windows. A prior uploaded-audio test was reported to work.

Lexical retrieval, not dense embeddings.

Current retriever is sparse TF-IDF with cosine similarity and intent-aware ranking. Evaluate neural embeddings or hybrid reranking if allowed, using the same gold evidence set.

Session state is in memory.

Restarting the app loses context; use durable, access-controlled session storage and stronger identity controls before production.

Demo and public repo pending.

No fabricated `demo.mp4` was added. Record a real <=120-second demo after direct-microphone/live provider checks; publish this local project to a public repo and verify access.

06 Fixes made

The current build adds browser microphone recording with WAV conversion, validated voice uploads, ticket CSV readback and lookup-first routing to avoid duplicate ticket creation. Regression tests cover these paths. Dataset validation, order verification and honest tool-result handling remain exercised by tests.

07 Reproduce locally

COMMANDS (FROM PROJECT ROOT)

```
pip install -r requirements.txt

# copy .env.example to .env

# set LLM_API_KEY, LLM_MODEL,

# GNANI_API_KEY, DATASET_DIR

python scripts/build_index.py

DATASET_DIR=<dataset-root> pytest -q

python scripts/evaluate_questions.py

python scripts/evaluate_conversations.py

uvicorn app:app --host 127.0.0.1 --port 8000

python run.py --input-dir sample_audio \

  --output-dir outputs
```

Links / submission status

GitHub: github.com/aayush-as69/voiceops-zentromart
Verify this source is pushed and public before review.
Demo MP4: pending target-browser and credentialed voice test.
SDK: github.com/Gnani-AI-Mintlify/gnani-vachana

Batch command

`python run.py --input-dir sample_audio --output-dir outputs` processes actual audio files; missing configuration exits clearly without mock substitutions.